

Contents

4. Success Metrics	1
6.1 Platform-Level Metrics	1
6.2 Application-Level Metrics	2
Usage	2
Governance	2
Data Quality	3
Query Quality	3
6.3 Metric Interpretation	3
Governance block rate	3
6.4 Review Cadence	4
6.5 Analytics Dashboard	4

4. Success Metrics

Metrics are captured from day one at both the platform level and the application level. Governance health metrics are first-class success indicators. A platform that is highly used but poorly governed is not successful.

Component definitions referenced in the metrics below (SMR, FQE, RAPL, SEG, NSE, Analytical Lineage Store (ALS)) are in Chapter 2 -- Core Platform Capabilities. Analytical Lineage Store (ALS) · Semantic Execution Governance · Narrative Synthesis Engine

6.1 Platform-Level Metrics

Metric	Definition	Target
Platform uptime	API availability (p99 end-to-end latency < 2s including FQE backend execution; error rate < 0.1%)	99.9%
Governance block rate	Queries blocked by governance checks ÷ total queries	Monitor — sustained > 15% warrants investigation; > 30% triggers mandatory configuration review (see §6.3)
FQE error rate	Queries with FQE execution errors ÷ total executed queries	< 2%

Metric	Definition	Target
Cache hit rate	Queries served from the FQE's internal result cache ÷ total executed queries. The result cache is an internal FQE component — this ratio is derived from the <code>cache_hit</code> field in the Analytical Lineage Store (ALS).	≥ 35% by month 2
Lineage completeness	Queries with complete lineage records ÷ total queries	100% — any deviation is a platform defect
SMR registry health	Metrics with no owner OR not updated in 12 months ÷ total active metrics	< 5%

6.2 Application-Level Metrics

Usage

Metric	Definition	Target
Weekly active users (WAU)	Distinct users executing at least one query in a 7-day window	50% of entitled users by day 90
Queries per user per week	Total query executions ÷ weekly active users	≥ 8
Drilldown rate	Sessions with at least one drilldown traversal ÷ total sessions	≥ 20%
Export rate	Sessions with at least one result export ÷ total sessions	≥ 15%
Narrative consumption	Queries where user expanded the narrative card ÷ queries with narrative	≥ 40%
Lineage inspector opens	Queries where user opened lineage inspector ÷ total queries (Power Analysts)	≥ 25%

Governance

Metric	Definition	Target
Metric resolution success rate	Queries where all requested metrics resolved from SMR ÷ total queries	≥ 95%
Cost circuit breaker rate	Queries blocked by cost limits ÷ total queries	< 5% — sustained > 10% triggers review
Classification gate block rate	Queries blocked by classification gate ÷ total queries	Monitored — any spike triggers security review

Metric	Definition	Target
Entitlement error rate	Queries with METRIC_NOT_ENTITLED or DIMENSION_NOT_ENTITLED errors ÷ total queries	< 3%
SMR approval backlog	Metric definitions in <code>proposed</code> or <code>in_review</code> state > 7 days	0 — all pending definitions reviewed within 7 days
Metric owner coverage	Active metrics with an assigned owner ÷ total active metrics	100%

Data Quality

Metric	Definition	Target
Engine error rate	Engine sub-plan failures ÷ total sub-plan executions (per engine)	< 1% per engine
Partial result rate	Queries returning partial results (timeout on one or more sub-plans)	< 2%
Stale data rate	Queries where a metric's data refresh cadence exceeded its SLA	< 5%
Cache freshness	Cached results served beyond TTL ÷ cache hits	< 1%

Query Quality

Metric	Definition	Target
Intent resolution accuracy	Queries where user accepted resolved intent without modification (when <code>requiresIntentConfirmation: true</code> is configured). When intent confirmation is not enabled, use query reformulation rate as the proxy measure.	≥ 85%
Query reformulation rate	Queries where user rephrased within 2 turns after a resolution error	< 10%
Narrative validation failure rate	Narrative synthesis attempts failing post-generation validation	< 2%

6.3 Metric Interpretation

Governance block rate

Block rate	Interpretation
< 5%	Healthy — some queries are appropriately blocked
5–15%	Review recommended — entitlement or scope configuration may need tuning

Block rate	Interpretation
15–30%	Likely misconfiguration — cost limits, entitlements, or scope too restrictive
> 30%	Configuration review mandatory — platform may be inaccessible to legitimate queries

Lineage completeness is measured as `completed_lineage ÷ total_queries`. Any value below 100% triggers an automated platform alert. Lineage gaps are platform defects, not acceptable operational variance.

SMR approval backlog counts metric definitions in `proposed` or `in_review` state for more than 7 calendar days. A non-zero count triggers a notification to the Application Admin.

6.4 Review Cadence

Cadence	Activity	Owner
Daily	Lineage completeness check; FQE error rate; classification gate spike detection	Platform Engineering (automated)
Weekly	WAU; governance block rate; SMR approval backlog; engine error rate per engine	Platform Engineering + Application Admin
Monthly	Full metric review; query quality analysis; SMR health report; narrative validation rate	Platform team + Application Admins
Day 90	WAU adoption assessment (50% target); drilldown adoption; export rate	Application Admin + Platform team
Quarterly	SMR completeness; metric owner coverage; entitlement policy review	Application Admin + Metric Owners

6.5 Analytics Dashboard

The Application Admin has access to a read-only dashboard showing:

- WAU trend (30-day rolling)
- Query volume by intent pattern
- Governance block rate trend with breakdown by circuit breaker type
- SMR metric resolution success rate
- Execution engine error rate per engine
- Cache hit rate trend
- Top 10 most-queried metrics
- Lineage completeness (always 100% or an active alert)
- SMR approval backlog count

Platform-level infrastructure metrics are visible to the Platform team only.